BEDO0001: PHYSICAL EDUCATION & YOGA

Objective: Students will learn the introduction of Physical Education, the Concept of fitness and wellness, Weight management and the lifestyle of an individual. The student will also learn about the relationship of Yoga with mental health and value education. In this course, students will also learn about the aspects of the Traditional games of India.

Module No.	Content
	1. **Physical Education:** Meaning, Aim, Need, Importance and Scope of Physical Education in the Modern Society. Physical Education in India before and after Independence. 2. **Concept of Fitness and Wellness:** Meaning, Importance, and Factors affecting Fitness and Wellness. Components and fitness equipment. 3. **Weight Management:** Meaning, Factor affecting weight management. BMI: (Body mass index) Meaning, charts, range and category. Obesity: Meaning, Causes and its types, Solutions for Overcoming Obesity. 4. **Lifestyle:** Meaning, Definition, Importance, Factor affecting Lifestyle. Healthy Lifestyle through Diet. Relationship between Diet and Fitness. 5. **Yoga and Meditation**: Definition, importance of Yoga. Yoga relation with mental health and sports. Definition of Asana and Pranayama, differences between Asana and physical exercise. Asana-Suraya- Namaskar, Bhujang Asana, Naukasana, Halasana,Vajrasan, Padmasana, Shavasana, Makrasana, Dhanurasana, TadAsana. Pranayam - Anulom, Vilom, Bhramary. 6. **Traditional Games of India:** Meaning, types and benefits of Traditional Games. 7. **Recreation in Physical Education:** Meaning, Importance, and Benefits of Recreation. Types of Recreational Activities, Aerobics and Zumba (Fit India Movement).

Chapter No - 3

Weight Management and Obesity

1. Meaning of weight management
2. Factor affecting weight management
3. BMI: (Body mass index) Meaning
4. Charts of BMI
5. Range and category

1. Meaning of weight management

Weight management is the process of making long-term lifestyle changes to maintain a healthy body weight.

Weight reduction is usually recommended for obese individuals, and for overweight persons with other health risk factors (such as, hypertension, hyperlipidemia, noninsulin-dependent diabetes, and family history of diabetes, women with a history of gestational diabetes or with a baby born large for gestational age). Weight reduction is also recommended for overweight individuals with heart disease, gout, gallbladder disease, and in situations where excessive weight imposes a functional burden (that is, chronic obstructive pulmonary disease, congestive heart failure, osteoarthritis of the spine, hips, or knees). Obese children and teens should avoid restrictive diets and weight loss efforts in favor of increased activity and moderation in food intakes, allowing them to grow into‖ their body weights, excessive television viewing can contribute to undesirable weight gains in young people. Although an overweight childhood or adolescence does not necessarily guarantee obesity later on in life, about 30% of obese adults developed the condition during childhood (before age 18), and 80% to 85% of obese teens eventually become obese adults. Obesity beginning early in life tends to involve a significant amount of fat cell hyperplasia, making it more difficult to control than excess weight gained later in life. Obesity is a multi factorial problem which has no single distinct cause. Only some 5% of obesity occurs secondary to a specific disease, such as a tumor of the hypothalamus, Cushing‘s syndrome, or Prader-Willis syndrome. Most obese conditions are due to an undetermined combination of individual genetic familial, environmental, physiological, and psychological factors.

2. Factor affecting weight management

Many factors can affect weight management, including:

- **Energy balance**: When you consume more calories than you burn through physical activity and daily life, your body stores the excess calories as fat.
- **Physical activity**: Regular physical activity helps maintain a healthy weight by improving metabolism, supporting muscle mass, and burning fat. The general recommendation for adults is to get 150 minutes of physical activity each week.
- **Diet**: Eating or drinking too many foods and beverages that are high in calories, sugar, and fat can lead to weight gain. Added sugar can change your body's hormones and biochemistry, contributing to weight gain.
- **Sleep**: Not getting enough good-quality sleep can affect weight management.
- **Mental health**: Mental ill-health can affect appetite and motivation to be active or eat healthy food.
- **Medications**: Some medicines can increase appetite or slow metabolism.
- **Environment**: Where you live, work, and play can affect your weight. For example, living near a grocery store with healthy, affordable food can make it easier to make healthier choices.
- **Genetics**: Family history and genes can affect weight.
- Cultural background.
- Medical conditions and disability.
- Drugs, Alcohol.
- Where people live, go to school and work.
- The media
- Time pressures - which can make it hard to cook healthy meals and exercise.
- Lifestyle.
- Shift work - due to disrupted sleep and eating patterns etc.

TIPS ON WEIGHT MANAGEMENT

1. Dieting is a life time commitment and not done on a crash basis.
2. For weight loss, Calories intake should not exceed 1500 cal./ day and should not go below 1000 cal. per day.
3. Total calories intake should be less than total calories expenditure per day to lose weight.
4. In order to do this, one should exercise lightly at least once or twice every day.
5. Exercises should be done after 1 to 1 hours alliterative meal.

6. Exercises should be both aerobic and Music forming.
7. Weight / measurement should be taken weekly at the same time every week.
8. All meals including breakfast should be eaten.
9. No snacks should be eaten in between.
10. If hungry in between meals, drinks water/eat fibrous foods.
11. Water may also be drunk during meals.
12. Avoid eating sugars, soft drinks, cakes etc.
13. Avoid too much dairy products including cream, butter and cheese.
14. Avoid fried food and oily foods.
15. Avoid alcohol.
16. Eat sufficient complex carbohydrates such as bread, potatoes, rice, chapattis, etc.
17. Eat sufficient green vegetables and fruits for roughage.
18. Preferably use a small plate and avoid 2nd helpings.
19. Maintain meal timings.
20. Enlist the support of your family and friends and remember that dieting is a life commitment and not a sporadic activity.
21. Eat a light-dinner, medium lunch, heavy breakfast.

3. BMI: (Body mass index) Meaning

Body mass index (BMI) is a calculation that estimates body fat and determines if someone is underweight, normal weight, overweight, or obese.

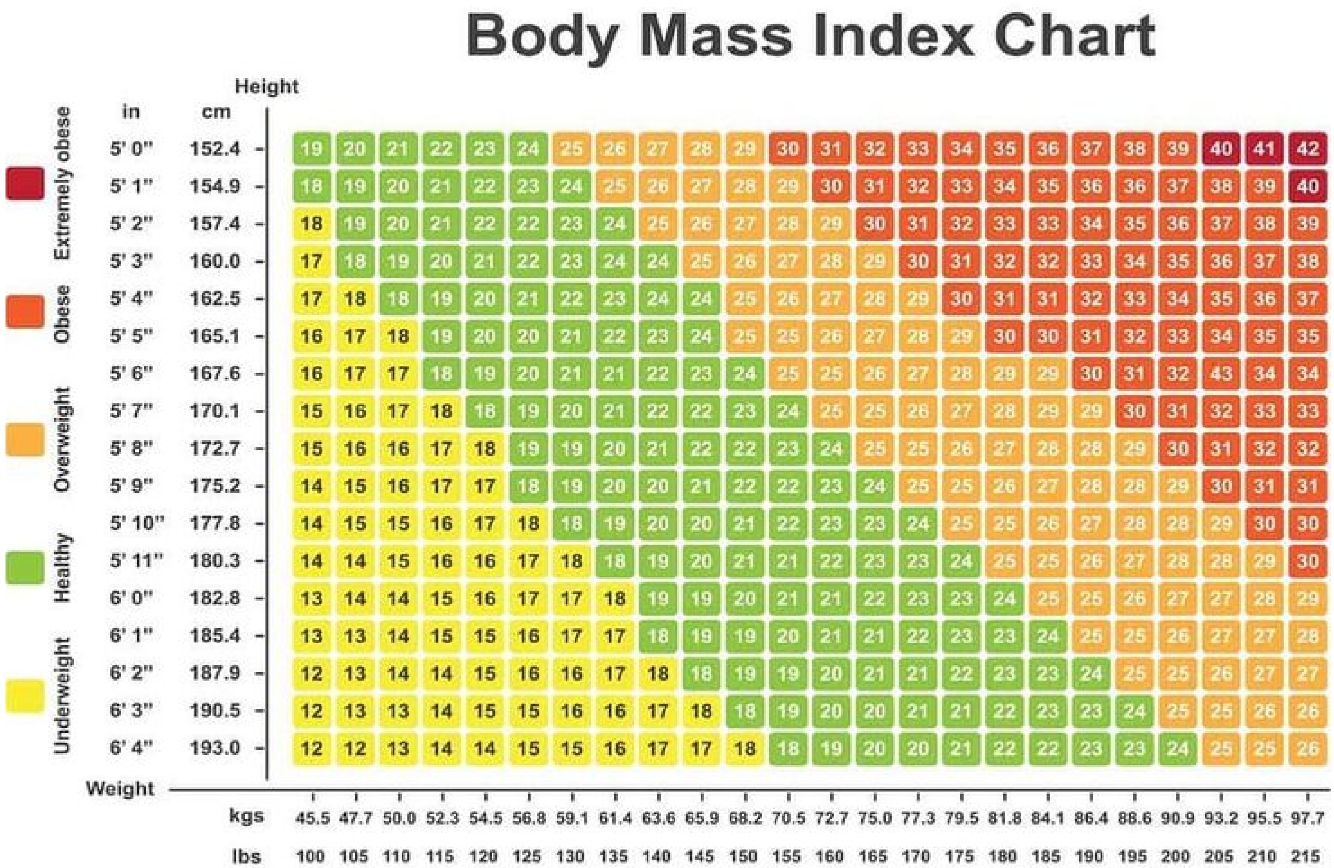
Obesity is defined as a condition with accumulation of excess body fat. Do you think that a measure of how much fat a person has in its body would serve as a tool for classification of obesity? No, because the measurement of direct body fat is difficult, so we use an indirect method, a ratio called the Body Mass Index (BMI) also termed Quetelet's index. This ratio exterminates dependence on frame size and provides the most useful method of measuring obesity in populations. BMI can be calculated from the following equation:

BMI = Weight (kg) / Height (m)2

Where kg = kilogram, m = metre

By this method, various grades of obesity, normal and underweight can be known. Our BMI value near 18.5 to 24.9 is the ideal value for us to remain healthy and enjoy a quality life. Table 9.1 presents the weight status according to the BMI range.

4. Charts of BMI



5. Range and category

WHO Adult BMI Categories

BMI-Range	Category
< 16.0	Severely Underweight / बहुत कम वजन
16.0 - 18.4	Underweight / कम वजन
18.5 - 24.9	Normal / सामान्य
25.0 - 29.9	Overweight / ज़्यादा वजन

30.0 - 34.9	Moderately Obese / मध्यम रूप से मोटा
35.0 - 39.9	Severely Obese / बहुत मोटा
> 40.0	Morbidly Obese / अत्यधिक मोटा

***Source: WHO**

So have you calculated the ratio for yourself? You can keep it a secret if you so desire.

Obesity

1. Meaning of Obesity
2. Causes and its types
3. Solutions for Overcoming Obesity

1. Meaning of Obesity

OBESITY

Obesity and overweight are not synonymous terms, yet most individuals who are 20% or more overweight are obese, and body weight at this level constitutes an established health risk. Individuals who weight 10% to 20% above the recommended range are not necessarily obese or over fat, since the body‘s weight may be largely composed of muscle tissues and / or bone. Some normal weight individuals are actually over fat, that is, the percent of body fat is undesirably high although the body weight is not above the recommended range.

Obesity is a state in which there is a generalized accumulation of excess adipose tissue in the body leading to more than 20 percent of the desirable weight. Obesity invites disability, disease and premature death. Excess body weight is a hindrance, leading to breathlessness on moderate exertion and predisposes a person to diseases like atherosclerosis, high blood pressure, stroke, diabetes, gall bladder diseases and osteoarthritis of weight bearing joints, varicose veins.

A certain amount of adipose, that is, fat tissue, is important for insulation of the body from heat, cold, and mechanical shock; to serve as a energy supply when glycogen reserves are depleted and food intake is delayed; and for protection against environmental stresses.

The obese are more prone to hypertension, dependent diabetes, arthritis, and other chronic diseases. Obese men have a higher mortality rate from cancers of the colon, rectum, and prostate; obese women from cancers of the gallbladder, breast (postmenopausal), uterus, and ovaries, obese individuals may have shorter than normal life spans.

Usually obesity is due to positive energy balance. That is, the intake of calories is more than the expenditure of calories.

Overweight and obesity are defined as abnormal or excessive fat accumulation that presents a risk to health. A body mass index (BMI) over 25 is considered overweight, and over 30 is obese.

In 2019, an estimated 5 million non-communicable disease (NCD) deaths were caused by higher-than-optimal BMI.

While obesity is a disease, it doesn't cause specific symptoms.

2. Causes and its types

On the most basic level, obesity happens when you consume more calories than your body can use. Many things may play a role in why you may eat more food than your body needs:

- **Genetic factors:** Genetic inheritance probably influences 50-70 percent a person's chance of becoming fat more than any other factor. A genetic base regulates species differences in body fat and sexual differences within a species. Within families the chance is 80 percent if both parents are obese, only 10 percent if neither parent is obese. A mutation in the human gene for the B3 receptor in adipose tissue, involved in lipolysin and thermo genesis markedly increase the risk of obesity.
- **Age and Sex:** It can occur at any age in either sex as long as the person is under positive energy balance.
- **Eating habits:** Certain eating habits may lead to obesity.

1. Nibbling between meals is common among housewives and is a potential cause for obesity.
2. Some may eat faster taking less time for chewing, therefore they tend to consume more food.
3. Obese persons respond to external cues to eat rather than internal hunger signals. They eat when it is mealtime or when they are surrounded by tasty foods instead of when they are hungry.
4. Housewives who are fond of cooking variety of foods or persons who are working in the kitchens may become obese.

5. Business executives who frequently attend business lunches have more chance of becoming obese.
6. Housewives who do not want leftover foods to be thrown out may consume forcibly and put on weight.
7. Some persons eat more food when they are unhappy as a compensation mechanism.

- **Physical activity:** Obesity is found in persons who lead sedentary lives and pay less importance to physical activity. Though obesity can occur at any age, this is more common during middle age when physical activity decreases without corresponding decrease in food consumption.
- **Certain medications**: Examples are Steroids, diabetes medications and Beta-Blockers.
- **Disability**: Adults and children with physical and learning disabilities are most at risk for obesity.
- Lack of physical activity
- Lack of sleep
- Stress
- Underlying health issues etc.

Assessment of obesity (Obesity Types)

There are several methods used worldwide to establish a correct weight, over weight and obesity. Such methods include:

- Height weight range.
- Brocas Index.
- Body mass index (BMI).
- Body fat measurement (skin fold)
- Others.

Each of these methods is indicative, and is to be used separately and individually, since they differ marginally from each other in conclusion. Therefore to come to an assessment, use any one method.

Body Mass Index In this method the weight in kgs is divided by the height in Mtrs x 2 – i.e. Wt in Kilogram

For example if an individual is 95 kilogram in weight and has a height of 1.80 mtrs, the calculation is - 95 kilograms / 1.80 x 2 (3.60 mtrs) = 26.38

Grading of BMI is as follows.

Not obese < 25 and below

Grade I obesity 25 – 29.9

Grade II obesity 30 – 40

Grade III obesity 40 > and above

What are the complications of obesity?

- **Cardiovascular diseases**: Having obesity increases your risk for cardiovascular diseases, including Coronary artery disease, Heart failure, Heart attack, and stroke.
- **Fatty liver disease:** Excess fats circulating in your blood make their way to your liver, which is responsible for filtering your blood. When your liver begins storing excess fat, it can lead to chronic liver inflammation and long-term liver damage.
- **Kidney disease:** High blood pressure, diabetes and liver disease are among the most common contributors to chronic kidney disease.

3. Solutions for Overcoming Obesity

A) LOW CALORIE METHOD

1. Weight

Your weight should conform to your height as per the height / weight chart. If necessary, you should reduce by a combination of exercises and diet.

2. Calories

Your maximum calorie intake should not exceed your current weight in kgs x 24. This is your Sedentary Metabolic rate.

a) If you are to lose weight, do an hour exercise daily and reduce your calorie intake of between 1000 – 1500 calories daily.

b) Do not reduce your calorie intake by more than 500 calories daily from your maximum intake e.g., (weight = 80 kgs, therefore maximum calorie intake = 80 x 24 = 1920; therefore, prescribed reduction = 1920 – 500 = 1420 calories).

c) Do not go below 1000 Calories ever.

d) When you achieve your correct weight, you consumption of calories should be correct weight x 24 and you must continue your daily exercise.

e) Do 1 hour 500 Calorie exercise output daily?

3. Diet

Sample diet plan of 1500 Calorie below

Sample diet India

Meal	Item	Total Calories
Bed Tea	1 cup tea with no sugar 20	20
Breakfast	2 Chappatis 1 katori Curry, OR 1 Masala dosa / 2 idlis	275
Lunch	1 cup Cooked rice, OR 3 Chappatis 1 cup Dal / Sambar 1 cup Curry 1 cup Salad / Curd 1 Fruit	600
Evening Tea	1 cup Tea with milk, no sugar	20
Dinner	1 cup Cooked rice, OR 3 Chappatis 1 cup Dal or Sambar 1 cup Curry 1 cup Salad / Curd	500
Nightcap	1 cup Milk	75

B) LOW FAT / HIGH FIBRE METHOD

High fibre a menu loaded with high-fibre foods is generally very filling yet low in calories. For instance, five apples have about the same caloric content as a small candy bar. High-fibre items also absorb a lot of water, which makes you feel full and they appear and feel bulkier too, which psychologically can make you think that you are eating more food than your actual caloric intake. Furthermore, fibre-rich foods such as carrots, raw spinach, and wholegrain breads require more chewing than lighter foods. They take longer to eat, which will help you eat less, too. Another major way that fibre appears to contribute to weight loss and overall health is through its absorption-delaying property. That a long-term high-fibre diet slows the absorption of an intake of glucose.

High fibre low calorie foods like green leafy vegetables, fruits, vegetable salads, whole grain cereals and pulses can be included in the diet. Inclusion of high fibre foods in diets for obese has many advantages. They are 1. Low in calorie density. 2. Foods like greens provide many vitamins and minerals (which are difficult to meet with restricted food) 3. Give satiety 4. Help in regulating bowel movements 5. Reduce blood cholesterol 6. Promote chewing and decreases rate of ingestion. Higher intake of fibre automatically cut down fat and calories.

British Nutrition Foundation (1990) has established the effectiveness of dietary fibre intake in achieving significant reducing in body weights without any side effects. Low fat high fibre method.

Limit fat (grams) to between 15-30 gms per day.

Eat at least 30-75 grams of fibre per day.

Drink at least 1 cup dairy food daily. (Skimmed milk, curd etc.)

Eat at least 2 cups of fresh fruit or vegetable daily.

Avoid refined sugars & alcohol.

Do not exceed 1500 calories; do not go below 1000 calories per day.

Utilize 500 calories per day for exercise (i.e. 1 hour of exercise).

C) HISTORICAL TREATMENTS FOR OBESITY

Recommendations for obesity treatment can be traced back to Ancient Greek and Roman times. Physicians around Hippocrates' time suggested that overweight individuals should 'reduce food and avoid drinking to fullness', and take regular exercise, particularly 'running during the night' and 'early morning walks'.

Several also recommended emetics and cathartics – some of the earliest weight-loss drugs. Emetics including hellebore plants and honey water were advised 'for the evacuation of the nourishment two or three times a month to all men and women'. Cathartics were composed, for example, of juice of scammony (bindweed), Canadian berry and sea spurge. Mild laxatives included donkey milk with honey, wild parsley, dodder of thyme (Cuscuta epithymum) and honey water or sweet wine. The writings of Aelianus (170–235 CE) reported the use of perhaps the earliest sleep apnoea treatment, whereby physicians 'prepared very long, thin needles and pushed them through the hips and belly of [Ancient Greek statesman] Dionysius when he had fallen into a deep sleep'.

Galen recorded treatments for obesity that included strenuous running every morning followed by a warm bath, a light meal and more physical work. Sushruta recommended physical work and described additional remedies for obesity such as 'fasting, exercise and depletory measures'. In the 1620s, British doctor Tobias Veneer first used the term 'obese' to describe people who were very overweight, and recommended bathing in the warm springs of

Bath ‘to make slender such bodies as are too gross’. The modern era of pharmacotherapy arguably began in the late 19th century, when preparations of thyroid hormone, extracted for use in the treatment of hypothyroidism, were also prescribed as anti-obesity therapy, largely due to their thermogenic properties. Early in the 20th century, obese individuals began to search for effective remedies, to reach the societal ideal of slimness; manufacturers preyed upon their naivete with a vast range of quack remedies. ‘Fatoff’ was marketed for overweight and obesity, despite containing the ineffective ingredients 10% soap and 90% water. Likewise, the product ‘Human-Ease’, marketed as a cure for obesity and ‘all known diseases’ was later found to consist of 95% lard.

- Asana, Pranayama and Meditation practice.